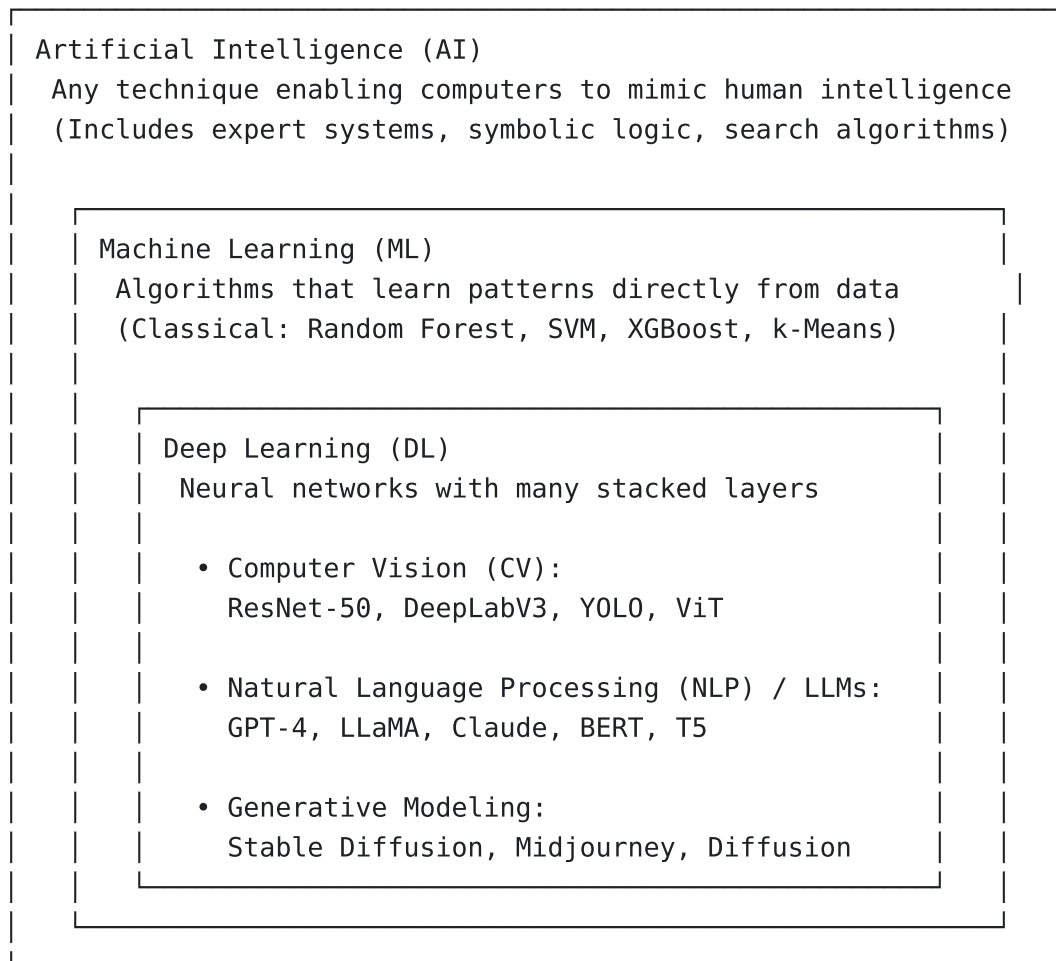


AI, Machine Learning, and Deep Learning: Taxonomy & Architecture Guide

A structured technical guide clarifying the conceptual hierarchy of Artificial Intelligence, the broader ecosystem of Machine Learning models, the core architectural divergence between **Convolutional Neural Networks (CNNs)** and **Transformers**, and the foundational role of optimization algorithms like **Backpropagation (BP)**.

1. The Core Hierarchy: Concentric Domains

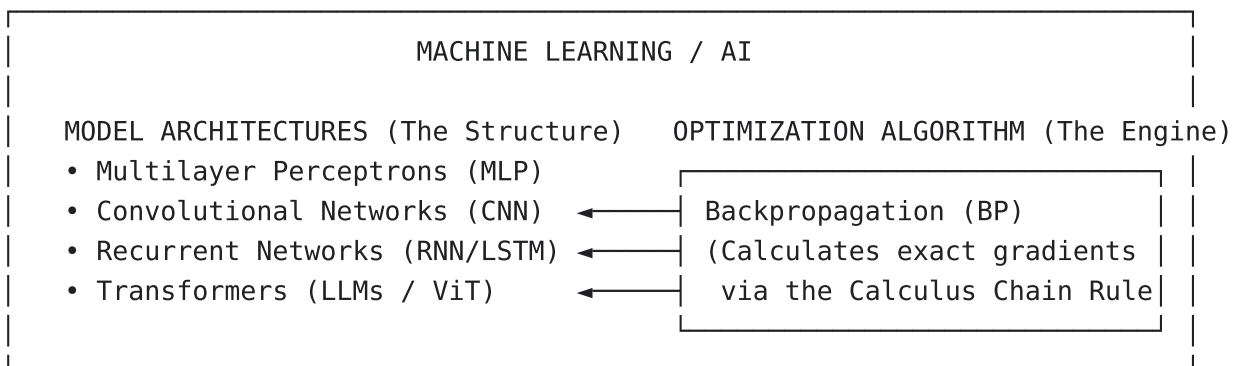
A common misconception is that Machine Learning and AI are mutually exclusive categories, or that modern LLMs represent "AI" while vision models represent "only ML." In computer science, they exist in a strict **subset hierarchy**:



- **All Machine Learning is AI**, but not all AI is Machine Learning (e.g., A^* pathfinding, chess engines using minimax with alpha-beta pruning, and rule-based expert systems are AI, but not ML).
- **All Deep Learning is Machine Learning** (and therefore also AI).
- **Both Computer Vision models (ResNet, DeepLabV3) and LLMs are Deep Learning**, meaning both belong to Machine Learning and both belong to AI.

2. Architectures vs. Learning Algorithms: Where Does Backpropagation (BP) Fit?

To understand how the **Backpropagation (BP)** algorithm relates to CNNs and AI, it is essential to distinguish between **Network Architectures** and **Optimization Algorithms**:



A. Does Backpropagation "Belong" to CNNs?

- **No, BP does not belong to CNNs.**
- Instead, **CNNs use Backpropagation.**
- Backpropagation is a universal training algorithm for artificial neural networks. It calculates the partial derivatives of the loss function with respect to every weight parameter ($\frac{\partial \mathcal{L}}{\partial W}$) using the **chain rule of calculus**.
- It applies equally to standard Multi-Layer Perceptrons (MLPs), Recurrent Neural Networks (RNNs via Backpropagation Through Time / BPTT), Convolutional Neural Networks (CNNs), and modern Transformers (LLMs).

B. Historical Milestones & The 1990s Era

- **1970–1974 (Mathematical Foundations):** Reverse accumulation automatic differentiation was discovered by Seppo Linnainmaa (1970) and applied to neural networks by Paul Werbos (1974).
- **1986 (The Breakthrough):** David Rumelhart, Geoffrey Hinton, and Ronald Williams published their landmark paper, *"Learning representations by back-propagating errors"*, demonstrating

that BP could train multi-layer neural networks to learn internal representations.

- **1989 & 1998 (Applying BP to CNNs):** Yann LeCun and collaborators applied backpropagation specifically to convolutional architectures to recognize handwritten zip codes (1989) and checks (the famous **LeNet-5** in 1998).
- **The Mid-1990s (1995–1996 Context):**
 - Around 1996, the neural network paradigm (known as **Connectionism** in AI) was fighting through a challenging period often associated with the late "AI Winter."
 - In 1995, Corinna Cortes and Vladimir Vapnik introduced **Support Vector Machines (SVMs)**, which came with convex optimization guarantees and temporarily overshadowed neural networks trained with BP.
 - Despite this, LeCun, Bengio, and Hinton persisted in using BP to train early CNNs through the 1990s, setting the stage for the deep learning revolution of 2012 (AlexNet).

C. Did BP in 1996 Belong to AI?

- **Yes, absolutely.** Backpropagation was, and remains, the core learning engine of **Connectionist AI** (neural network-based AI) and Machine Learning.
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3. Where ResNet-50 and DeepLabV3 Fit

ResNet-50 and DeepLabV3 belong to the domain of **Computer Vision (CV)**, powered by **Deep Learning**:

- **ResNet-50 (Residual Network, 50 layers):** A Convolutional Neural Network (CNN) architecture designed for image classification and visual feature extraction. It introduced residual skip connections ($y = \mathcal{F}(x, \{W_i\}) + x$) to solve the vanishing gradient problem during backpropagation in very deep networks.
- **DeepLabV3:** A semantic segmentation architecture designed to assign a semantic category to every individual pixel in an image. It uses atrous (dilated) convolutions and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale context without sacrificing spatial resolution.

Because these models learn weights via backpropagation over large datasets (such as ImageNet or COCO), they are **Deep Learning models**, which places them squarely inside **Machine Learning** and **Artificial Intelligence**.

4. Where LLMs (Large Language Models) Fit

LLMs belong to the domain of **Natural Language Processing (NLP)** and **Generative Modeling**, powered by **Deep Learning**:

- **Transformer Architecture:** Modern LLMs (e.g., LLaMA, GPT, Gemini, Mistral) are deep neural networks based on the multi-head self-attention mechanism introduced by Vaswani et al.

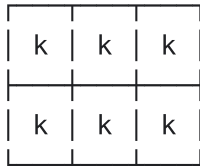
(2017).

- **Learning Mechanism:** Just like ResNet, LLMs are trained using gradient descent and **backpropagation** on vast datasets (optimizing next-token cross-entropy loss instead of image classification loss).
 - **Relationship to ML:** LLMs are not fundamentally distinct from Machine Learning; they are simply scaled-up deep neural sequence models.
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5. Deep Architectural Dive: CNNs vs. Transformers

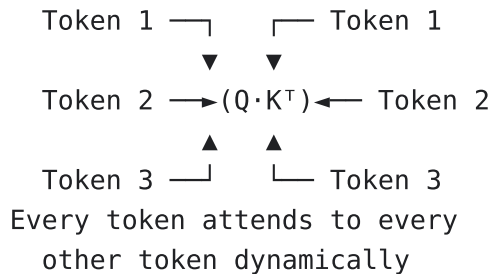
The core operational difference between models like ResNet-50 / DeepLabV3 and modern LLMs lies in their **underlying architectural primitives** and how they represent and process input tensors.

CONVOLUTIONAL (CNN)
[Local Spatial Kernels]



Sliding across 2D Grids
(Translation Invariance + Locality)

TRANSFORMER (ATTENTION)
[Global Pairwise Attention]



A. Convolutional Architectures (ResNet-50, DeepLabV3)

- **Input Structure:** Continuous 3D/4D spatial tensors with shape (B, C, H, W) —Batch, Channels (e.g., RGB), Height, and Width.
- **Core Operation (Discrete 2D Convolution):**

$$(I * K)(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n)$$

- **Inductive Biases:**
 - **Locality:** Pixels close to each other are strongly correlated. Filters only look at small patches (e.g., 3×3 or 7×7) at a time.
 - **Translation Equivariance:** A feature (like an edge, eye, or wheel) is recognized regardless of where it appears in the image coordinate plane.
- **Receptive Field:** Starts very narrow in early layers and grows hierarchically as layers are stacked. Global scene context is only achieved deep within the network.

B. Transformer Architectures (LLMs & Vision Transformers)

- **Input Structure:** 2D/3D sequential embeddings with shape (B, N, D) —Batch, Sequence Length (N tokens), and Embedding Dimension (D).

- **Core Operation (Scaled Dot-Product Self-Attention):**

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Where Query (Q), Key (K), and Value (V) are linear projections of the input sequence tokens.

- **Inductive Biases:**
 - **Extremely Weak Inductive Bias:** Transformers do not inherently assume locality or sequential order; they require explicit **Positional Encodings** to know which token comes first, second, or last.
 - **Dynamic Global Receptive Field:** From the very first layer, any token can directly attend to and interact with any other token across the entire context window.

C. The Convergence: Vision Transformers (ViT)

Transformers are no longer restricted to text. The **Vision Transformer (ViT)** applies pure Transformer architectures directly to image data:

1. An image of shape (H, W, C) is sliced into a grid of non-overlapping patches (e.g., 16×16 pixels).
2. Each patch is flattened into a 1D vector and linearly projected into an embedding dimension, acting exactly like a "word token."
3. The sequence of patch tokens is processed by standard Transformer encoder blocks with self-attention.
4. **Outcome:** ViT outperforms CNNs when trained on massive datasets (hundreds of millions of images) because its low inductive bias allows it to learn broader patterns without being constrained by fixed 3×3 convolutional filters.

6. Architectural Comparison Matrix

Dimension	Convolutional Networks (CNNs)	Transformer Architectures
Exemplar Models	ResNet-50, DeepLabV3, ConvNeXt, YOLO	GPT-4, LLaMA-3, BERT, ViT, Swin
Primary Native Data	Grid / Tensor data (Images, Audio spectrograms)	Sequential tokens (Text, Code, DNA, Patches)
Computational Mechanism	Fixed-weight sliding spatial kernels ($3 \times 3, 5 \times 5$)	Pairwise dot-product attention ($\text{Softmax}(QK^T / \sqrt{d})V$)
Receptive Field Growth	Linear / gradual (requires deep stacking or dilation)	Instantaneous / global (all-to-all from Layer 1)

Dimension	Convolutional Networks (CNNs)	Transformer Architectures
Inductive Bias	High (hardcoded spatial locality & translation equivariance)	Low (learns arbitrary pairwise relationships from scratch)
Data Efficiency	High on small-to-moderate datasets	Low on small datasets; excels exponentially at massive scale
Computational Complexity	Linear with image resolution: $\mathcal{O}(H \cdot W)$	Quadratic with sequence length: $\mathcal{O}(N^2)$ (unless using FlashAttention/linear variants)

7. The Broader Landscape of Machine Learning Models

Machine Learning encompasses a vast spectrum of algorithms beyond deep neural networks. These are broadly categorized by learning paradigm and architectural family:

A. Classical Supervised Learning

Used extensively in industry for tabular data, forecasting, and low-latency prediction:

- **Tree-Based Ensembles:**
 - **Random Forest:** Ensemble of decision trees trained with bagging to reduce variance.
 - **Gradient Boosted Decision Trees (GBDT):** Algorithms like **XGBoost**, **LightGBM**, and **CatBoost** that sequentially minimize residual errors. Often superior to neural networks on tabular datasets.
- **Linear & Kernel Models:**
 - **Support Vector Machines (SVM):** Finds optimal hyperplanes in high-dimensional feature spaces.
 - **Ridge / Lasso Regression:** Linear models with L_2 and L_1 regularization for continuous prediction and feature selection.
 - **Logistic Regression:** Probabilistic linear classifier for binary and multi-class classification.

B. Unsupervised & Dimensionality Reduction

Algorithms that discover latent structures without labeled target variables:

- **Clustering:** k -Means, DBSCAN, Hierarchical Clustering, Gaussian Mixture Models (GMM).
- **Dimensionality Reduction:** Principal Component Analysis (PCA), t-SNE, UMAP.
- **Anomaly Detection:** Isolation Forests, One-Class SVM.

C. Deep Learning Architectural Families

Modern deep architectures tailored to specific modalities:

- **Convolutional Neural Networks (CNNs):** ResNet, EfficientNet, ConvNeXt (Images, Video, Audio spectrograms).
- **Recurrent Architectures:** LSTM, GRU, Mamba/State-Space Models (Sequential data, Time-series).
- **Transformers:** BERT (encoder-only), GPT/LLaMA (decoder-only), T5 (encoder-decoder), Vision Transformers (ViT).
- **Diffusion & Generative Models:** Stable Diffusion, DALL-E, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs).

8. Model Taxonomy Reference Matrix

Model / Technique	Primary Domain	Core Family	Is It ML?	Is It AI?	Primary Use Case
Backpropagation (BP)	General Neural Optimization	Optimization Algorithm	Yes	Yes	Computing gradients to update neural network weights
ResNet-50	Computer Vision	Deep Learning (CNN)	Yes	Yes	Image classification, feature extraction backbone
DeepLabV3	Computer Vision	Deep Learning (CNN + ASPP)	Yes	Yes	Semantic pixel-level segmentation
XGBoost / LightGBM	Tabular / Structured	Classical ML (GBDT)	Yes	Yes	Risk scoring, fraud detection, tabular prediction
Random Forest	Tabular / Multi-modal	Classical ML (Ensemble Trees)	Yes	Yes	Classification, regression, feature importance

Model / Technique	Primary Domain	Core Family	Is It ML?	Is It AI?	Primary Use Case
Support Vector Machine (SVM)	Text / Tabular	Classical ML (Kernel methods)	Yes	Yes	High-dimensional classification
LLaMA / GPT-4	Language / Multimodal	Deep Learning (Transformer)	Yes	Yes	Text generation, reasoning, code synthesis
Vision Transformer (ViT)	Computer Vision	Deep Learning (Transformer)	Yes	Yes	Patch-based image classification and representation
Stable Diffusion	Computer Vision / Art	Deep Learning (Latent Diffusion)	Yes	Yes	Text-to-image generation
A* Search / Minimax	Search & Planning	Symbolic / Heuristic AI	No (Rule/Search)	Yes	Pathfinding, game playing (non-learning)

9. Summary & Key Takeaways

- AI encompasses ML, which encompasses Deep Learning ($DL \subset ML \subset AI$):** ResNet, DeepLabV3, and LLMs all belong to both Machine Learning and AI.
- Backpropagation (BP) is an algorithm, not an architecture:** BP does not belong exclusively to CNNs; it is the fundamental calculus-based optimization technique used to train MLPs, CNNs, RNNs, and Transformers alike.
- CNNs rely on strong spatial priors:** By enforcing local sliding kernels, models like ResNet-50 excel at processing grid-based pixel tensors with high parameter efficiency and translation invariance.
- Transformers rely on global pairwise attention:** By dropping spatial assumptions, models like GPT and ViT allow every element (whether a word token or an image patch) to interact with every other element, scaling dramatically with large data and compute.
- The modalities are converging:** Vision Transformers (ViT) and Multimodal LLMs now use the Transformer attention mechanism directly on visual patch tokens, merging vision and language

under a unified architectural paradigm.